Method of Moments-type Estimation of Dependence and Marginal Distributions in Dependently Censored Survival Data

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Introduction

Background

- In bivariate competing risks survival data where only the minimum of the time-to-events is observed and never both, dependence between the survival endpoints is non-identifiable.
- If dependence exists between the time-to-events, cause-specific hazards analysis under the independent censoring assumption becomes biased.
- The current study estimates the correlation (strength of dependence) in bivariate competing risks or dependently censored survival data T = time to event of interest and C = time to competing event or dependent censoring by a method of moments-type parametric approach.

Copulas for dependence modeling in survival analysis

- Copula: A multivariate CDF for which the marginal distribution of each random variable is the Uniform distribution on the interval [0, 1].
  Ex) $C_\theta(S_1(x), S_2(x))$ with copula parameter θ: A bivariate joint survival probability of event types 1 and 2, where the correlation or strength of dependence is determined by θ
- Archimedean copula: If a generator function φ(·) can be used to create a copula of the form $\varphi^{-1}\{\varphi(u) + \varphi(v)\}$, then the copula is Archimedean.
  Ex) Clayton, Frank, and Gumbel copulas
- Zheng and Klein (1994, 1995) defined a copula-graphic estimator for dependent competing risks data and showed that the marginal survival functions are identifiable under an "assumed" copula dependence structure.
- Recently, Deresa and Van Keilegom (2019) and Czado and Van Keilegom (2021) formulated a "parametric" identifiability approach in dependently censored survival data.

Proposed Method

Study notations and settings

- (T, C) of sample size n can be competing risks data or dependently censored survival data with parametric marginal distributions for a time-to-event of interest T and time-to competing event or dependent censoring C. (T, C) have a correlation ρ_0 through a parametric copula.
- Sample mean data of $(\bar{T}, \bar{C})$, with $\bar{T} = \bar{t}_m$ (with mean $\mu_{\bar{t}_m}$, variance $\sigma^2{}_{\bar{t}_m}$) or missing (·), $\bar{C} = \bar{c}_{n-m}$ (with mean $\mu_{\bar{c}_{n-m}}$, variance $\sigma^2{}_{\bar{c}_{n-m}}$) or missing (·), where $m = \sum_{i=1}^{n} I(T \le C)$, $\bar{T} = \frac{1}{m}\sum_{i=1}^{n} T_i \cdot I(T \le C)$, $\bar{C} = \frac{1}{(n-m)}\sum_{i=1}^{n} C_i \cdot \{1 - I(T \le C)\}$, and $(\bar{T}, \bar{C})$ have the same correlation ρ_0 as in (T, C) through a Gaussian copula by the bivariate central limit theorem.
- A hypothetical bivariate parametric distribution (Y_1, Y_2) of sample size n with parameters vector θ for which $\min(Y_1, Y_2)$ and $I(Y_1 \le Y_2)$ produce the same sample mean information as $(\bar{T}, \bar{C})$ above, i.e. $\bar{Y}_1 = \bar{y}_{1,m} = \bar{t}_m$ (with mean $\mu_{\bar{t}_m}$, variance $\sigma^2{}_{\bar{t}_m}$) or missing (·), $\bar{Y}_2 = \bar{y}_{2,n-m} = \bar{c}_{n-m}$ (with mean $\mu_{\bar{c}_{n-m}}$, variance $\sigma^2{}_{\bar{c}_{n-m}}$) or missing (·), where $m = \sum_{i=1}^{n} I(Y_1 \le Y_2)$, $\bar{Y}_1 = \frac{1}{m}\sum_{i=1}^{n} Y_{1,i} \cdot I(Y_1 \le Y_2)$, $\bar{Y}_2 = \frac{1}{(n-m)}\sum_{i=1}^{n} Y_{2,i} \cdot \{1 - I(Y_1 \le Y_2)\}$.

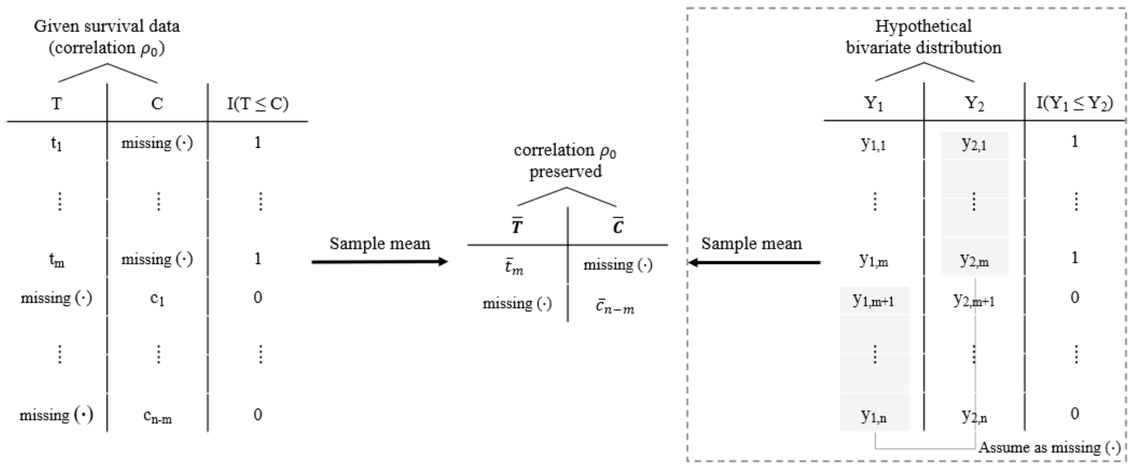

Figure 1. A hypothetical bivariate parametric distribution (right) with the same sample mean information as that of a given competing risks or dependently censored survival data (left)

Conjectures and the estimation procedure

- Conjecture 1: A parametrically identifiable bivariate distribution that produces the same sample mean information as that of the given (T, C) has the same correlation ρ_0 of interest.
- Conjecture 2: The correlation parameter of the hypothetical bivariate distribution can be numerically estimated with the given survival data's sample mean information of $\frac{m}{n}$, $\mu_{\bar{t}_m}$, $\mu_{\bar{c}_{n-m}}$, $\sigma^2{}_{\bar{t}_m}$, and $\sigma^2{}_{\bar{c}_{n-m}}$ through a method of moments-type estimation procedure.
- Estimation procedure:

Obj. function
- Objective function to minimize aggregated sample mean (partial moments) comparison error between the given data (T, C) and the hypothetical bivariate distribution (Y_1, Y_2) as in Figure 1.
- Ex) For a bivariate Weibull distribution with 5 parameters, $f_{obj}(\cdot) = (\sum_{i=1}^{5} w_i \times |\varepsilon_i|)$ and the objective is to find $\hat{\theta}$ that minimizes the aggregated error such that $\hat{\theta} = argmin_{\theta \in \Theta}(f_{obj}(\cdot))$.

Param. space
- Calculate lower and upper bounds of the parameter vector initial values to be entered into the objective function, using copula-graphic estimation of marginal distributions under four candidate correlations (none: 0, low: 0.3, moderate: 0.5, high: 0.8).
- The R packages *copula*, *compound.Cox* were used.

Global search
- Global search of the parameter space using simulated annealing under four candidate correlations (0, 0.3, 0.5 or 0.8). A bootstrap aggregation (bagging) algorithm determines which candidate correlation is selected via voting among B bootstrap samples of the given data (T, C).
- The R package *GenSA* was used.

Local search
- Local refinement of the correlation parameter estimated in the global search above as either 0, 0.3, 0.5 or 0.8. The correlation parameter now allowed to vary within [0, 1).
- The R functions *optim()*, *nlminb()* were used.

Precision
- Calculate the precision of the correlation estimate using B bootstrap samples as:
  ① Arrange correlation estimates in increasing order $\hat{\tau}_{(1)} \le \hat{\tau}_{(2)} \le \cdots \le \hat{\tau}_{(B)}$
  ② Calculate the $100 \cdot (1-\alpha)$ percentiles confidence interval $\left[\hat{\tau}_{\left(B \cdot \frac{\alpha}{2}\right)}, \hat{\tau}_{\left(B \cdot \left(1-\frac{\alpha}{2}\right)\right)}\right]$

Figure 2. Estimation procedure of the correlation (strength of dependence) in a given competing risks or dependently censored survival data

Simulation study

Data generation

- The conditional CDF method to generate a correlated data pair (u, v) using copula C(u, v):
  Using $c_u(v) = \frac{\partial}{\partial u} C(u, v) = Pr(V \le v \mid U = u)$,
  ① Generate instances of two independent Uniform(0,1) random variables u and t.
  ② Obtain v using $c_u(v) = t$ and $v = c_u^{-1}(t)$.
- Generate correlated parametric survival times (t, c) from the correlated data pair (u, v):
  For marginal CDFs $F_T(t)$, $G_C(c)$, obtain t and c as $t = F_T^{-1}(u)$ and $c = G_C^{-1}(v)$.

Parameter settings

- Marginal distribution parameter settings for each bivariate competing risk or dependently censored survival data simulated were
  ① Bivariate Exponential: $T \sim \text{Exp}(\lambda_T = 0.025)$, $C \sim \text{Exp}(\lambda_C = 0.039)$
  ② Bivariate Weibull: $T \sim Weibull(\lambda_T = 0.06,\ \alpha_T = 0.63)$, $C \sim Weibull(\lambda_C = 0.04,\ \alpha_C = 0.86)$
  ③ Bivariate Log-Normal: $T \sim \text{LN}(\mu_T = 2.2,\ \sigma_T = 1.0)$, $C \sim \text{LN}(\mu_C = 2.0,\ \sigma_C = 0.25)$
  ④ Weibull and Log-Normal: $T \sim Weibull(\lambda_T = 0.066,\ \alpha_T = 1.1)$, $C \sim \text{LN}(\mu_C = 2.0,\ \sigma_C = 0.25)$

Simulation results

- Simulation results of correlation (Kendall's tau) estimation with the proposed method in the simulated data (T, C) and its 200 bootstrap samples are shown in Table 1.
- Simulation results of correlation (Kendall's tau) estimation with the proposed method in 100 instances of simulated data (T, C) with 100 bootstrap samples each are shown in Table 2.
- Sample sizes were 200 or 500 for Kendall's tau of 0 or 0.8, and 1000, 1500, or 2000 for Kendall's tau of 0.3 or 0.5.
- The Clayton copula was used as the underlying copula linking the two marginal distributions.
- Abbreviations: CI, confidence interval; CP, coverage probability; MAE, mean absolute error; SE, standard error

Marginal distributions	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Point estimate	Bootstrap MAE	Bootstrap SE	Bootstrap 95% CI
Bivariate Exponential	0	0.088	0.007	0.024	(-0.099, 0.245)
	0.3	0.374	0.000	0.050	(0.206, 0.398)
	0.5	0.497	0.005	0.040	(0.413, 0.615)
	0.8	0.696	0.039	0.033	(0.586, 0.863)
Bivariate Weibull	0	0.005	0.043	0.077	(-0.067, 0.186)
	0.3	0.249	0.042	0.085	(0.127, 0.471)
	0.5	0.450	0.031	0.085	(0.347, 0.679)
	0.8	0.720	0.015	0.033	(0.694, 0.824)
Bivariate Log-Normal	0	-0.070	0.064	0.081	(-0.108, 0.207)
	0.3	0.266	0.022	0.045	(0.188, 0.381)
	0.5	0.483	0.023	0.032	(0.389, 0.552)
	0.8	0.825	0.059	0.071	(0.628, 0.912)
Weibull and Log-Normal	0	-0.093	0.055	0.053	(-0.095, 0.109)
	0.3	0.276	0.076	0.059	(0.154, 0.360)
	0.5	0.524	0.054	0.049	(0.402, 0.558)
	0.8	0.732	0.063	0.057	(0.660, 0.845)

Table 1. Simulation results of correlation estimation in 200 bootstrap samples of dependently censored survival data (T, C)

Marginal distributions	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Mean estimate	MAE	Empirical SE	CP
Bivariate Exponential	0	0.028	0.068	0.088	96
	0.3	0.295	0.061	0.077	97
	0.5	0.523	0.061	0.070	96
	0.8	0.805	0.062	0.075	95
Bivariate Weibull	0	0.061	0.077	0.081	96
	0.3	0.262	0.069	0.075	92
	0.5	0.558	0.074	0.068	96
	0.8	0.815	0.066	0.077	97
Bivariate Log-Normal	0	0.013	0.063	0.083	98
	0.3	0.287	0.055	0.060	94
	0.5	0.492	0.054	0.049	96
	0.8	0.751	0.068	0.030	96
Weibull and Log-Normal	0	0.021	0.068	0.082	93
	0.3	0.286	0.070	0.090	92
	0.5	0.493	0.053	0.068	94
	0.8	0.752	0.067	0.072	97

Table 2. Simulation results of correlation estimation in 100 multiple runs of 100 bootstrap samples of dependently censored survival data (T, C)

Conclusion

- The simulation results show the good performance of the proposed method in terms of the point estimate and its bootstrap SE (Table 1), the mean estimate and MAE (Table 2), as well as the CPs being near the desired 95% under a 0.05 significance level.
- The current study proposed a novel method of estimating the correlation in dependently censored survival data which does not require the presence of informative covariates.